

Distractor-Based Distractors: Adventures in Assessment Item Analysis

Russell Jeter Darryl Chamberlain Jr. Kelvin Rozier
Georgia State University Embry-Riddle Aeronautical University Georgia State University

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Motivation and Context

This poster presents a methodology for capturing underlying student conceptions based on their answers to a series of multiple-choice questions. We quantitatively and qualitatively examine publisher-supplied assessment items across four exams in College Algebra.

Item-Level Assessment Analysis

To appropriately capture students' mathematical conceptions, we perform an item-level analysis utilizing the following statistical measures: Item Difficulty (ID), Rasch Logistic Model (RLM) Estimated Item Difficulty (EID), and Point-Biserial Correlation (PBC). Item Difficulty (ID) refers to the proportion of students that correctly answered a given item. The PBC coefficient measures the correlation between students' performance on a single item and their performance on the exam as a whole. The RLM is a mathematical log function that relates the probability of a (correct) response on an item to the person's mathematical ability and to the item's difficulty. In this model, students have the highest probability of answering items with EID near their estimated ability (Krabbe, 2017). Unlike PBC, RLM takes into account a student's performance on each item in relation to the rest of the class' performance on the item. Example statistics are presented in Table 1 for two items on the exam.

Table 1. Item Response Analysis for two assessment items. Both items have similar difficulty, that is, Item Difficulty and RLM Estimated Item Difficulty are both consistent, but 4C22 is predictive of student performance on the exam (PBC close to 0.5), while 4C10 is not (PBC close to zero).

IRT Analysis			
<u>Item</u>	<u>RLM EID</u>	<u>PBC</u>	<u>ID</u>
4C10	2.91	0.03	0.36
4C22	2.75	0.38	0.39

Distractor Identification and Analysis

To investigate differences in item-level statistics, we investigate items' distractors – plausible but incorrect solutions (Chamberlain Jr. & Jeter, 2024). This leads to a classification of distractor types that can capture students' content misconceptions and provides evidence for distractor type impact on item-level statistics.

Discussion and Conclusions

These analyses and the items of interest they point to raise the following questions: What do these distractors tell us about student thinking? Why do high performing students get these questions right and low performing students get them wrong? To extend a multiple-choice assessment to fill a more formative role, it is important to align choices of distractors with the ways students might be thinking. Right/wrong dichotomy of questions in the analysis limits the amount of predictive power when it comes to student thinking.

References

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